

Sample question:

Suppose that Alice's RSA public key is $(N, e) = (33, 3)$ and her private key is $d = 7$.

a. If Bob encrypts the message $M = 19$ using Alice's public key, what is the ciphertext C ? Show that Alice can decrypt C to obtain M .

Answer:**🔒 RSA Encryption & Decryption**

Encryption: $C = M^e \bmod N$

Decryption: $M = C^d \bmod N$

📧 Encryption:

$$\begin{aligned} C &= M^e \bmod N \\ &= 19^3 \bmod 33 \\ &= 6859 \bmod 33 \\ &= 28 \end{aligned}$$

👉 **$C = 28$**

🔒 Decryption:

$$\begin{aligned} M &= C^d \bmod N \\ &= 28^7 \bmod 33 \\ &= 28^3 * 28^3 * 28 \bmod 33 \\ &= 7 * 7 * 28 \bmod 33 \\ &= 19 \end{aligned}$$

✅ Final Answer:

$$M = 19$$

Let $p=11$ and $q=13$. Compute the modulus N , Euler's totient $\phi(N)$, and find a valid public key exponent e such that $\gcd(e, \phi(N))=1$.

Given:

$$p = 11, q = 13$$

Step 1: Compute Modulus N

$$N = p \times q = 11 \times 13 = 143$$

👉 **$N = 143$**

Step 2: Compute Euler's Totient $\phi(N)$

$$\phi(N) = (p-1)(q-1) = (11-1)(13-1) = 10 \times 12 = 120$$

$$\phi(N) = 120$$

Step 3: Choose Public Exponent e

Condition:

$$\gcd(e, \phi(N)) = 1$$

☞ এমন একটা e নিতে হবে যেটা 120 এর সাথে co-prime

Possible values:

- $e = 7$ ✓ (কারণ $\gcd(7, 120) = 1$)
- $e = 11$ ✗ ($\gcd \neq 1$)

☞ তাই আমরা নেই:

$$\checkmark e = 7$$

Final Answer

$$\checkmark N = 143$$

$$\checkmark \phi(N) = 120$$

$$\checkmark \text{Valid public exponent: } e = 7$$

☞ Public Key হবে: $(N, e) = (143, 7)$

Co-Prime ber korar technique:

$$\gcd(e, 120) = 1$$

☞ মানে: e এবং 120 co-prime হতে হবে

◆ **Step 1: 120 এর prime factor বের করি**

$$120 = 2^3 \times 3 \times 5$$

☞ তাই e 2, 3, 5 দিয়ে divisible হওয়া যাবে না

◆ **Step 2: Possible e values (small numbers থেকে)**

চেক করি:

- 2 ✗ (2 দিয়ে যায়)
- 3 ✗ (3 দিয়ে যায়)
- 4 ✗ (2 আছে)
- 5 ✗
- 6 ✗
- 7 ✓
- 8 ✗
- 9 ✗
- 10 ✗
- 11 ✓
- 12 ✗
- 13 ✓

Using the RSA public key (N=143,e=7), encrypt the message M=10. Show all steps

Given:

N = 143, e = 7

M = 10

🔑 Formula:

$C = M^e \bmod N$

☑ Step 1: Compute 10^7

$10^7 = 10000000$

📁 **Step 2: Mod 143 নাও**

Better way (shortcut using modular):

$$10^2 = 100 \bmod 143$$

$$10^3 = 100 \times 10 = 1000 \equiv 1000 - 7 \times 143 = 1000 - 1001 = -1 \equiv 142$$

$$10^4 = (-1) \times 10 = -10 \equiv 133$$

$$10^5 = 133 \times 10 = 1330 \equiv 1330 - 9 \times 143 = 1330 - 1287 = 43$$

$$10^6 = 43 \times 10 = 430 \equiv 430 - 3 \times 143 = 430 - 429 = 1$$

$$10^7 = 1 \times 10 = 10$$

🏆 **Final Answer:**

Ciphertext C = 10

RSA Decryption Using the private key $d=103$ and modulus $N=143$, decrypt the ciphertext $C=100$. Show the complete calculation.

Solve koro

Diffie-Hellman Key Exchange Alice and Bob use the Diffie-Hellman key exchange with $p=23$ and $g=5$. If Alice selects her private key $a=6$ and Bob selects $b=15$, compute the shared secret key K .

Given:

$$p = 23, g = 5$$

$$\text{Alice: } a = 6$$

$$\text{Bob: } b = 15$$

◆ Step 1: Alice computes

$$A = g^a \mod p = 5^6 \mod 23$$

$$5^2 = 25 \equiv 2 \pmod{23}$$

$$5^4 = 2^2 = 4$$

$$5^6 = 5^4 \cdot 5^2 = 4 \cdot 2 = 8$$

👉 $A = 8$

◆ Step 2: Bob computes

$$B = g^b \mod p = 5^{15} \mod 23$$

$$5^2 = 2, \quad 5^4 = 4, \quad 5^8 = 16$$

$$5^{15} = 5^8 \cdot 5^4 \cdot 5^2 \cdot 5 = 16 \cdot 4 \cdot 2 \cdot 5$$

$$16 \cdot 4 = 64 \equiv 18$$

$$18 \cdot 2 = 36 \equiv 13$$

$$13 \cdot 5 = 65 \equiv 19$$

👉 $B = 19$

◆ Step 3: Shared Secret Key

Alice computes:

$$K = B^a \mod p = 19^6 \mod 23$$

$$19 \equiv -4$$

$$(-4)^2 = 16$$

$$(-4)^4 = 16^2 = 256 \equiv 3$$

$$(-4)^6 = 3 \cdot 16 = 48 \equiv 2$$

👉 $K = 2$

Bob checks:

$$K = A^b \mod p = 8^{15} \mod 23$$

(করলেও same result আসবে)

🎯 **Final Answer:**

👉 Shared Secret Key $K = 2$

Knapsack Encryption Given the superincreasing knapsack (2,3,7,14), a modulus $n=31$, and a multiplier $m=5$, compute the general knapsack.

Given:

Superincreasing knapsack: (2, 3, 7, 14)

$n = 31, m = 5$

🔒 **Formula:**

$$W' = W \times m \mod n$$

◆ **Step-by-step calculation:**

$$2 \times 5 = 10 \mod 31 = 10$$

$$3 \times 5 = 15 \mod 31 = 15$$

$$7 \times 5 = 35 \bmod 31 = 4$$
$$14 \times 5 = 70 \bmod 31 = 8$$

☞ General Knapsack (Public Key)

(10, 15, 4, 8)

Knapsack Decryption Using the general knapsack (10, 15, 5, 8) and ciphertext C=23, decrypt the message to retrieve the binary sequence.

✓ Method 2: Systematic (Largest → Smallest approach)

Knapsack: (10, 15, 5, 8)

☞ আগে **descending order** এ সাজাও:

(15, 10, 8, 5)

Target: 23

Step-by-step:

☞ **1. 15 নাও**

$$\text{Remaining} = 23 - 15 = 8$$

☞ **2. 10 নেওয়া যাবে?**

না ($8 < 10$) ✗

☞ **3. 8 নাও**

$$\text{Remaining} = 8 - 8 = 0 \checkmark$$

☞ কাজ শেষ (remaining = 0)

এখন mapping back করো original order এ:

Original: (10, 15, 5, 8)

- $10 \rightarrow 0$
- $15 \rightarrow 1$
- $5 \rightarrow 0$
- $8 \rightarrow 1$

🔗 Final Answer:

Binary sequence = 0101

39. ECC

A digital signature provides for data integrity and a MAC provides for data integrity.

Why does a digital signature also provides for non-repudiation while a MAC does not?

Consider the elliptic curve $E : y^2 = x^3 + 7x + b \pmod{11}$.

- Determine b so that the point $P = (4,5)$ is on the curve E .
- Using the b found in part a, list all points on E .
- Using the b found in part a, find the sum $(4,5) + (5,4)$ on E .
- Using the b found in part a, find the point $3(4,5)$.

Sample question:

Suppose that for the knapsack cryptosystem, the super increasing knapsack is (3,5,12,23) with $n = 47$ and $m = 6$.

- Give the public and private keys.
- Encrypt the message $M = 1110$ (given in binary). Give your result in decimal.

Answer:

□ **Given:**

🔗 Superincreasing knapsack (SIK):
(3, 5, 12, 23)

🔗 $n = 47, m = 6$

🔗 Message: **$M = 1110$ (binary)**

🔗 **(a) Public Key & Private Key**

🔒 Step 1: Private Key

📝 Rule:

Private key =

✓ $SIK + m + n$

📝 তাই:

✓ **Private Key = (3, 5, 12, 23), m = 6, n = 47**

🔒 Step 2: Public Key বের করি

📝 Formula:

$W' = W \times m \bmod n$

এক এক করে করি:

📝 $3 \times 6 = 18 \bmod 47 = 18$

📝 $5 \times 6 = 30 \bmod 47 = 30$

📝 $12 \times 6 = 72 \bmod 47 = 25$

📝 $23 \times 6 = 138 \bmod 47 = 44$

✓ **Public Key = (18, 30, 25, 44)**

🔒 (b) Encryption

🔒 Step 3: Binary message বুঝো

📝 $M = 1110$

মানে:

Bit Weight

1 18

1 30

1 25

0 44

☞ শুধু যেগুলো 1 আছে সেগুলো যোগ করো:

$$= 18 + 30 + 25$$

$$= 73$$

✓ **Final Answer:**

✓ (a)

- Private Key: (3, 5, 12, 23), $m = 6$, $n = 47$
- Public Key: (18, 30, 25, 44)

✓ (b)

☞ **Ciphertext** = 73 (decimal)

◆ What is Cryptography?

Cryptography is the practice of transforming information (messages) so that **only authorized people can understand it**.

◆ Main Functions:

- **Encryption:** Convert **plaintext** → **ciphertext**
- **Decryption:** Convert **ciphertext** → **plaintext**

◆ Purpose of Cryptography:

1. **Confidentiality:** Keep the information secret
2. **Integrity:** Ensure data is not altered
3. **Authentication:** Verify the sender's identity
4. **Non-repudiation:** Prevent the sender from denying sending the message

📎 Knapsack Problem কী?

Knapsack problem হলো একটি **subset sum problem**।

☞ Definition:

দেওয়া থাকে কিছু weight:

$W_0, W_1, \dots, W_{n-1}$
এবং একটি sum S

What is RSA?

RSA is a **public-key cryptography algorithm**, which means it uses **two keys**:

1. **Public Key** – for encryption (anyone can use it)
2. **Private Key** – for decryption (only the owner can use it)

It's called the **gold standard** of public-key crypto because it's widely used for secure communication.